Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

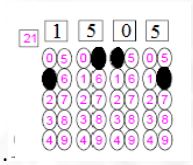
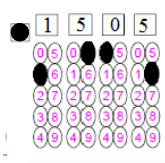
ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 12-01-2026**
Time: 02.00Pm to 05.00Pm **GTM-13** **Max. Marks: 300**

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black** Ball Point **Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75 Questions.** The maximum marks are **300.**
5. There are **three** parts in the question paper 1,2,3 consisting of **Mathematics, Physics** and **Chemistry** having **25 Questions** in each subject and subject having **two sections.**
   (I) Section –I contains **20 Multiple Choice Questions** with only one correct option.
   Marking scheme: +4 for correct answer, **0** if not attempt and **-1** in all other cases.
   (II) Section-II contains **05 Numerical Value Type Questions.**
   - The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

   To cancel any attempted question bubble on the question number box.
   For example: To cancel attempted Question 21. Bubble on 21 as shown below

Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, **0** if **not attempt** and **-1** in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold of make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number: | | | | | | | | | |

Candidate's Signature:_____ **Invigilator's Signature:** _____

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12-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Main _GTM-13_Test Syllabus

MATHEMATICS : Total Syllabus

PHYSICS : Total Syllabus

CHEMISTRY : Total Syllabus

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MATHEMATICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. If the circles $(x-1)^2+(y-3)^2=r^2$ and $x^2+y^2-8x+2y+8=0$ intersect in two distinct points, then

1) $2<r<8$ **2)** $r=2$ **3)** $r<2$ **4)** $r>2$

2. The number of solutions of $|[x]-2x|=4$ where [x] is the greatest integer $\le x$ is

1) 2 **2)**4 **3)**1 **4)** Infinite

3. A line makes angles $\alpha,\beta,\gamma,\delta$ with the four diagonals of a cube, then $\cos^2\alpha+\cos^2\beta+\cos^2\gamma+\cos^2\delta=$

1) $\frac{1}{3}$ **2)** $\frac{2}{3}$ **3)** $\frac{1}{5}$ **4)** $\frac{4}{3}$

4. Let 'S' be the set of lines belonging to family of lines $(1+\lambda)y+(\lambda-1)x+2(1-\lambda)=0$, then the length of the segment which belongs to S and makes shortest intercept on $y^2=4x-4$ is

1) 3 **2)** 4 **3)** 1 **4)** 2

5. The locus of the vertices of the family of parabolas $y=\frac{a^3x^2}{3}+\frac{a^2x}{2}-2a$ is

1) $xy=\frac{3}{4}$ **2)** $xy=\frac{105}{64}$ **3)** $xy=\frac{64}{105}$ **4)** $xy=\frac{35}{16}$

6. Four consecutive natural numbers are chosen from first 50 natural numbers. What is the probability that their sum is divisible by 3?

1) $\frac{15}{{}^{50}C_4}$ **2)** $\frac{15}{47}$ **3)** $\frac{{}^{16}C_4}{{}^{50}C_4}$ **4)** $\frac{3\times{}^{16}C_4}{{}^{50}C_4}$

7. If ω is a complex cube root of unity, then a root of the equation $\begin{vmatrix} x+1 & \omega & \omega^2 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix}=0$ is

1) x=0 **2)** x=1 **3)** $x=\omega$ **4)** $x=\omega^2$

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8. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set

1) is increased by 2 **2)** is decreased by 2

3) is two times the original median **4)** remains the same as that of the original set

9. If the inverse trigonometric functions take principal values, then

$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right)+\frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ equal to

1) 0 **2)** $\frac{\pi}{4}$ **3)** $\frac{\pi}{3}$ **4)** $\frac{\pi}{6}$

10. $\hat{a}$ *and* $\vec{b}$ are two non parallel vectors. The angles of the triangle, two sides of which are represented by $\sqrt{3}\left(\hat{a}\times\vec{b}\right)$ *and* $\vec{b}-\left(\hat{a}.\vec{b}\right)\hat{a}$

1) $30°, 60°$ and $90°$ **2)** $15°, 75°$ and $90°$

3) $20°, 60°, 100°$ **4)** $45°, 55°, 80°$

11. Let R_1 and R_2 be two relations defined on R by $aR_1b \Leftrightarrow ab \geq 0$ and $aR_2b \Leftrightarrow a \geq b$ then

1) R_1 is equivalence relation but not R_2

2) Neither R_1 nor R_2 is an equivalence relation

3) R_2 is an equivalence relation but not R_1

4) Both R_1 and R_2 are equivalence relations

12. Match the following List-I with List-II

Column-I	**Column-II**
(P) The domain of the function $f(x)=\sqrt{4x-3}+\sqrt{2x-6}$ is	(1) $\left[0,\frac{1}{2}\right]$
(Q) The range of the function $y=\frac{x^2}{1+x^4}$ is	(2) $[3,\infty)$
(R) The domain of $f(x)=\frac{\log_2(x+3)}{x^2+3x+2}$ is	(3) $\left(-\infty,\frac{2}{3}\right]\cup(1,\infty)$
(S) The range of $f(x)=\frac{x^2-2}{x^2-3}$	(4) $(-3,\infty)-\{-1,-2\}$

1) $(P)\to(1),(Q)\to(2),(R)\to(4),(S)\to(3)$ **2)** $(P)\to(3),(Q)\to(4),(R)\to(2),(S)\to(1)$

3) $(P)\to(4),(Q)\to(1),(R)\to(2),(S)\to(3)$ **4)** $(P)\to(2),(Q)\to(1),(R)\to(4),(S)\to(3)$

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13. Statement –I : The area of ellipse $2x^2+3y^2=6$ bounded with axes will be more than the area bounded by $2|x|+3|y|\le 6$ bounded with axes

Statement –II : The length of major axis of the ellipse $2x^2+3y^2=6$ is less than the distance between the points of $2|x|+3|y|\le 6$ on X-axis.

In the light of above statements, choose the correct answer from the options given below.

1) Both statement I and statement II are false

2) Both statement I and statement II are true

3) Statements I is true but statement II is false

4) Statement I is false but statement II is true

14. The area (in sq. units) of the largest rectangle ABCD whose vertices A and B lie on the x-axis and vertices C and D lie on the parabola, $y=x^2-1$ below the x-axis, is :

1) $\frac{2}{3\sqrt{3}}$ **2)** $\frac{1}{3\sqrt{3}}$ **3)** $\frac{4}{3}$ **4)** $\frac{4}{3\sqrt{3}}$

15. Let f be any function continuous on [a, b] and twice differentiable on (a,b) and twice differentiable on (a,b). If for all $x\in(a,b)$, $f'(x)>0$ and $f''(x)<0$, then for any $c\in(a,b)$, $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than :

1) $\frac{b+a}{b-a}$ **2)** 1 **3)** $\frac{b-c}{c-a}$ **4)** $\frac{c-a}{b-c}$

16. Set A has 3 elements and set B has 4 elements. The number of injective functions that can be defined from A to B is

1) 144 **2)** 12 **3)** 24 **4)** 64

17. $\frac{1}{3.6}+\frac{1}{6.9}+\frac{1}{9.12}+-----\text{to n terms}=$

1) $\frac{1}{n+1}$ **2)** $\frac{1}{9(n+1)}$ **3)** $\frac{n}{9(n+1)}$ **4)** $\frac{n}{n+1}$

18. The number of six letters words (with or without meaning) formed using all the letters of the word VOWELS so that all the consonants never come together is

1) 470 **2)** 576 **3)** 370 **4)** 501

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19. If A and B are symmetric matrices, then AB-BA is a

1) Symmetric matrix **2)** Skew-symmetric matrix

3) Diagonal matrix **4)** Null matrix

20. If the $2x+3y+z=0,\ 3x+y-2z=0$ equation and ax+2y-bz=0 has non-trivial solution, then

1) a-b = 2 **2)** a+b+1=0 **3)** a+b=3 **4)** a-b-8=0

SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

21. Let $f(x)=x^3+x^2f'(1)+xf''(2)+f'''(3),\ x\in R$. Then $f'(10)$ is equal to _____.

22. The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latus rectum to the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$, is

23. If $u_n=\int_0^{\pi/4}\tan^n x\,dx$, then $\frac{1}{u_5+u_3}=$ _____.

24. If the sixth term in the expansion of $\left(\frac{1}{x^{8/3}}+x^2\log_{10}x\right)^8$ is 5600. The value of x is _____.

25. The equation of a curve whose slope at any point is thrice its abscissa and which passes through (-1, -3) is $2y=\lambda(x^2-3)$, then the value of λ must be equal to_____.

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PHYSICS **Max Marks: 100**

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

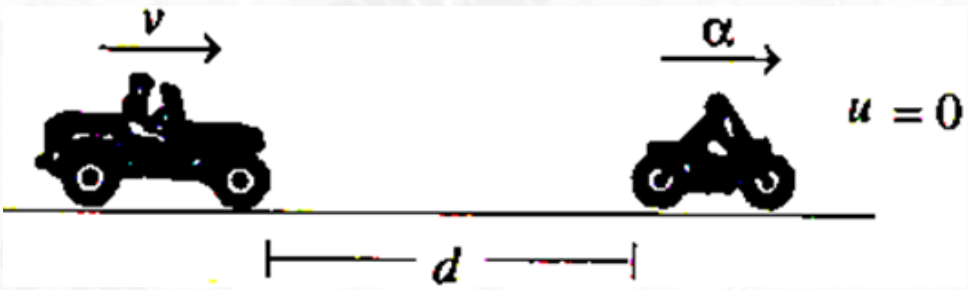
26. The electron volt (e.v) is the unit of:

1) Power **2)** Voltage **3)** potential **4)** Energy

27. The resultant of two forces 2P and $\sqrt{2}\,P$ is $\sqrt{10}\,P$ The angle between the forces is ?

1) 30^0 **2)** 60^0 **3)** 45^0 **4)** 90^0

28. A police inspector in a jeep is chasing a pickpocket on a straight road. The jeep is going at its maximum speed v (assumed uniform). The pickpocket rides on a waiting motorcycle when the jeep is a distance d away, and the motorcycle starts with a constant acceleration α. The pickpocket will be caught if



1) $v = 2\alpha d$ **2)** $v \geq \sqrt{2\alpha d}$ **3)** $v \geq \sqrt{2\frac{d}{\alpha}}$ **4)** None of these

29. The equation of trajectory of a projectile is $y = 10x - \frac{5}{9}x^2$. If we assume g = 10 ms^{-2} the range of projectile (in meters) is

1) 36 **2)** 24 **3)** 18 **4)** 9

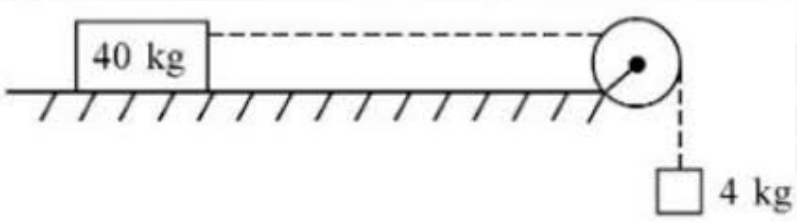
30. A rocket with a lift-off mass 20000 kg is blasted upwards with an initial acceleration of 5.0 m/s^2. Calculate the initial thrust (force) of the blast.

1) $3\times10^5\,N$ **2)** $6\times10^5\,N$ **3)** $4\times10^5\,N$ **4)** $5\times10^5\,N$

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31. A block of mass 40kg slides over a surface, when a mass of 4kg is suspended through an inextensible massless string passing over frictionless pulley as shown below. The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is. (Given $g = 10ms^{-2}$.)



1) 0.5 m/s^2 **2)** 0.3 m/s^2 **3)** 1 m/s^2 **4)** 0.7 m/s^2

32. The equation of motion of a projectile is $y = ax - bx^2$, where a and b are constants of motion. Match the quantities in Column I with the relations in Column II. (g=acceleration due to gravity)

Column I	Column II
(A) The initial velocity of projection	(P) a/b
(B) The horizontal range of projection	(q) $a\sqrt{\frac{2}{bg}}$
(C) The maximum vertical height attained by projectile	(r) $\frac{a^2}{4b}$
(D) The time of flight of projectile	(s) $\sqrt{\frac{g\left(1+a^2\right)}{2b}}$

1) A-p, B-q, C-r D-s **2)** A-s, B-p, C-q D-r

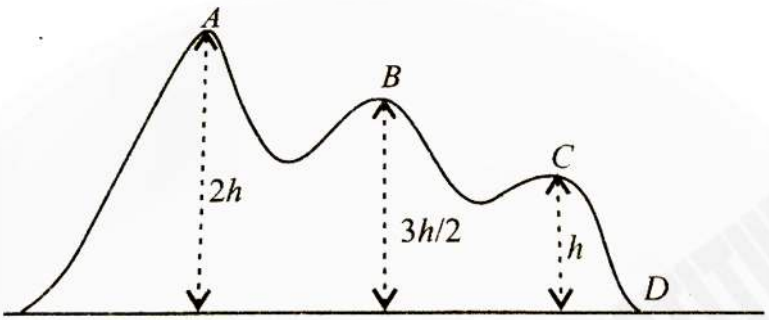
3) A-s, B-p, C-r D-q **4)** A-p, B-s, C-r D-q

33. A drunkard walking in a narrow lane takes 5 steps forward and 3 steps backward, followed again by 5 steps forward and 3 steps backward, and so on. Each step is 1 m long and requires 1 sec. How long the drunkard takes to a fall in a 13m away from the start?

1) 30 s **2)** 37 S **3)** 40 S **4)** 60 S

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JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHAITANYA REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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34. A small roller coaster starts at point A with a speed μ on a curved track as shown in the figure.



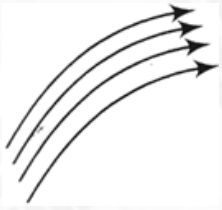
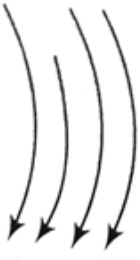
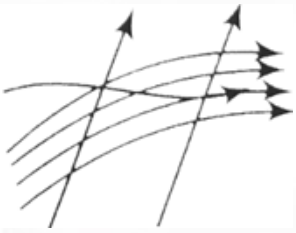
The friction between the roller coaster and the track is negligible and it always remains in contact with the track. The speed of roller coaster at point D on the track will be

1) $\left(\mu^2+gh\right)^{1/2}$ **2)** $\left(\mu^2+2gh\right)^{1/2}$ **3)** $\left(\mu^2+4gh\right)^{1/2}$ **4)** μ

35. Which of the following is most elastic?

1) Steel **2)** Rubber **3)** Glass **4)** Sponge

36. Which of the following diagrams (figure) does not represent a streamline flow?

1)  **2)**  **3)**  **4)** 

37. Statement-I: The total translational kinetic energy of all the molecules of a given mass of an ideal gas is 1.5 times the product of its pressure and its volume.

Statement-II: The molecules of a gas collide with each other and the velocities of the molecules change due to the collision.

1) Both (I) and (II) are true and (ii) is the correct explanation of (I)

2) Both (I) and (II) are true but (II) is not the correct explanation of (I)

3) (I) is true but (II) is false

4) (I) is false but (II) is true

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38. A wire stretched between two rigid supports vibrates in its fundamental mode with a frequency of 45 Hz. The mass of the wire is $3.5\times10^{-2}kg$ and its linear mass density is $4.0\times10^{-2}kg/m$. What is the tension in the string?

1) 250 N **2)** 208 N **3)** 300 N **4)** 248 N

39. Assertion: When two coils are wound on each other, the mutual induction between the coils is maximum.

Reason: Mutual induction does not depend on the orientation of the coils.

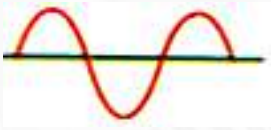
1) Both A and R are true and R is the correct explanation of A

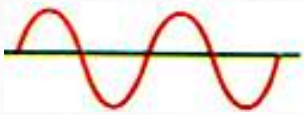
2) Both A and R are true and R is not the correct explanation of A

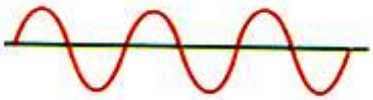
3) A is true but R is false

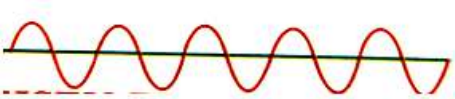
4) A is false but R is true

40. The de Brogile wave present in fifth Bohr orbit is:

1) 

2) 

3) 

4) 

41. As the intensity of incident light increases

1) Kinetic energy of emitted photoelectrons increase

2) photoelectric current decreases

3) photoelectric current increases

4) kinetic energy of emitted photoelectrons decreases

42. Statement-I: When a body acquires negative charge, its mass decreases.

Statement-II: A body acquires negative charge when it loses electrons.

1) Both Statement I and Statement II are false

2) Both Statement I and Statement II are true

3) Statement I is false but Statement II is true

4) Statement I is true Statement II is false

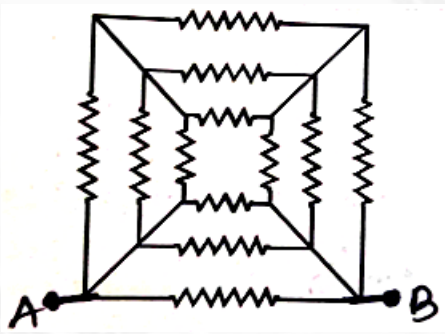
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43. A parallel beam of light of intensity I_0 is incident on a glass plate, 25 % of light is reflected by upper surface and 50% of light is reflected from lower surface. The ratio of maximum to minimum intensity in interference region of reflected rays is

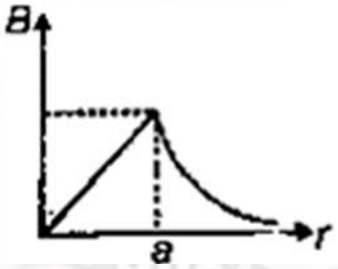
1) $\left(\frac{\frac{1}{2}+\sqrt{\frac{3}{8}}}{\frac{1}{2}-\sqrt{\frac{3}{8}}}\right)^2$ **2)** $\left(\frac{\frac{1}{4}+\sqrt{\frac{3}{8}}}{\frac{1}{2}-\sqrt{\frac{3}{8}}}\right)$ **3)** 5/8 **4)** 8/5

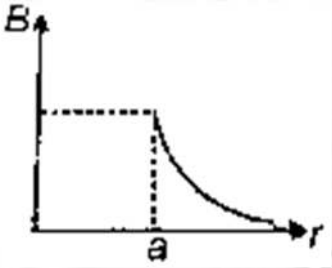
44. Twelve resistance each of resistance 16Ω are connected in the circuit as shown. The net resistance between AB is

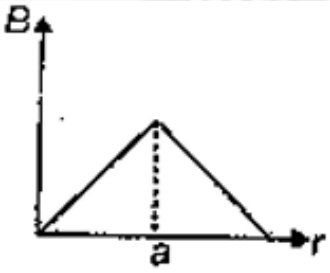


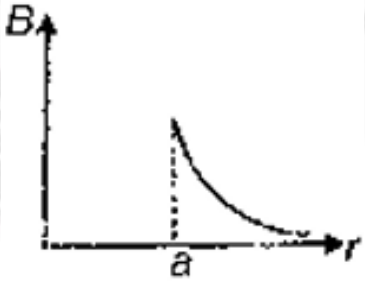
1) 1Ω **2)** 2Ω **3)** 3Ω **4)** 4Ω

45. The magnetic field due to a conductor for unifrom cross section of radius and carrying a steady current is represented by

1)


2)


3)


4)


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SECTION-II
(NUMERICAL VALUE TYPE

This section contains **5 Numerical Value Type Questions.** The Answer should be within **0 to 9999.** If the Answer is in **Decimal** then round off to the **Nearest Integer** value (Example i,e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11).**

Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

46. A current $i = 2\sin(\pi t/3)$ amp is flowing in an inductor of 2 henry. The amount of work done in (in joule) increasing the current from 1.0 amp to 2.0 amp is _____.

47. A new system of units is evolved in which the values of μ_0 and $\in_0$ are 2 and 8 new system units respectively. Then the speed of light(m/s) in this system will be $x\times10^{-2}$ new system units. Find x.

48. A radioactive substance contains 10000 nuclei and its half life period is 20 days. The number of nuclei present at the end of 10 days is (Take $2^{-0.5} = 0.707$)

49. A magnet of magnetic moment 5 A m^2 is freely suspended in a uniform magnetic field of strength 2T. Find the work done in rotating the magnet through an angle of 60^0 is given by_____J.

50. The value of current in the following diagram will be_____ mA



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CHEMISTRY

Max Marks: 100

SECTION-I
(SINGLE CORRECT ANSWER TYPE)

This section contains **20 Multiple Choice Questions.** Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

51. The set representing the correct order of ionic radii is

1) $Li^{+}>Be^{2+}>Na^{+}>Mg^{2+}$ **2)** $Li^{+}>Na^{+}>Mg^{2+}>Be^{2+}$

3) $Mg^{2+}>Be^{2+}>Li^{+}>Na^{+}$ **4)** $Na^{+}>Li^{+}>Mg^{2+}>Be^{2+}$

52. Which of the following statement is incorrect?

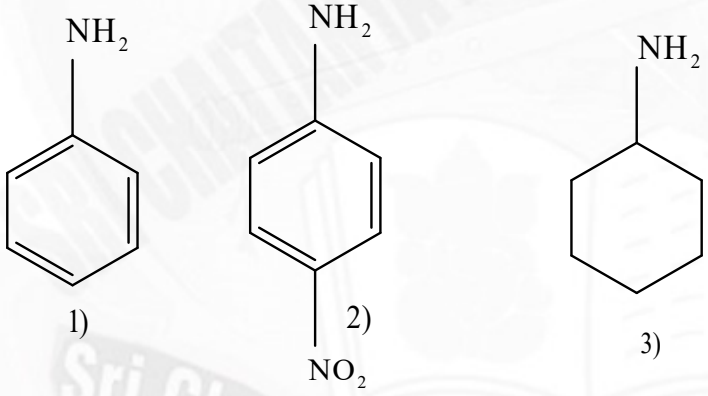
1) The order of splitting energy is $PtCl_4^{2-} > PdCl_4^{2-} > NiCl_4^{2-}$ (consider only magnitude)

2) $\left[Ni(CO)_4\right]$ is diamagnetic where as $\left[Ni(H_2O)_6\right]^{2+}$ is paramagnetic

3) $\left[Ni(CN)_4\right]^{2-} \rightarrow dsp^2$ hybridized

4) The magnetic moment of $K_3\left[Fe(CN)_6\right]$ is $\sqrt{8}\,B.M$

53. The correct order of basicity of the following compounds.

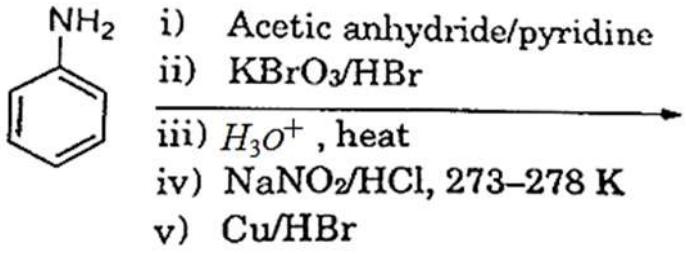


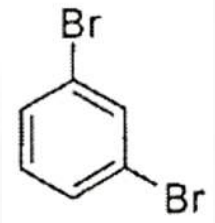
1) 2) 3)

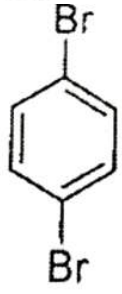
1) 2>1>3 **2)** 1>2>3 **3)** 3>1>2 **4)** 3>2>1

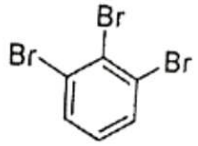
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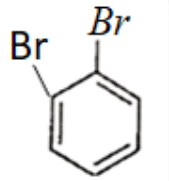
54. The product of the following reaction sequence is



1) 

2) 

3) 

4) 

55. The value of K_C is 64 at 800K for the reaction $N_2(g)+3H_2(g) \rightleftharpoons 2NH_3(g)$

The value of K_C at 800 K for the following reaction is: $NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g)+\frac{3}{2}H_2(g)$

1) $\frac{1}{8}$ **2)** 8 **3)** $\frac{1}{64}$ **4)** 64

56. Consider 'n' is the number of lone pair of electrons present in the equatorial position of the most stable structure of ClF_3. The ions from the following with ' n ' number of unpaired electrons are :

a) V^{3+} b) Ti^{3+} c) Cu^{2+} d) Ni^{2+} e) Ti^{2+}

Choose the correct answer from the options given below :

1) a and c only **2)** a, d and e only **3)** b and c only **4)** b and d only

57. For the reaction mechanism, of the reaction $2\,NO(g)+2H_2\,(g) \rightarrow N_2(g)+2H_2O(g)$.

$$Step\ I\ : 2NO \xrightleftharpoons{K_{eq}} N_2O_2 \quad ; \quad (fast)$$

$$Step\ II\ : N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O \quad ; \quad (slow)$$

$$Step\ III\ : N_2O + H_2 \xrightarrow{k_3} N_2 + H_2O \quad ; \quad (fast)$$

Expression of rate of reaction is

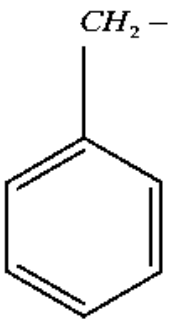
$(Take\ K_{eq} \times k_2 = k')$

1) $k'[NO]^2[H_2]$ **2)** $k'[N_2O_2][H_2]$ **3)** $k'[N_2O][H_2]$ **4)** $k'[N_2O_2]$

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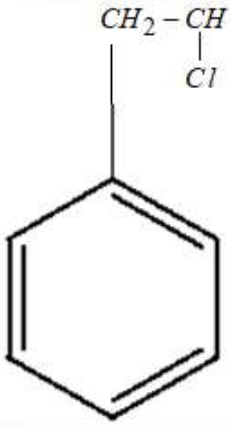

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58.

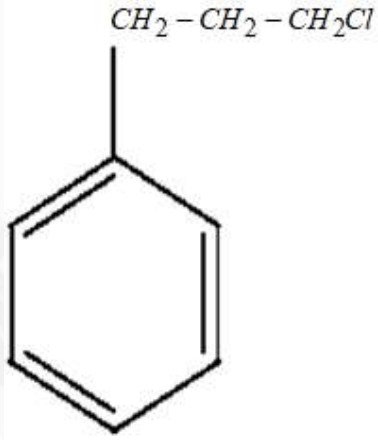


$+HCl \rightarrow X, X$ *is* :

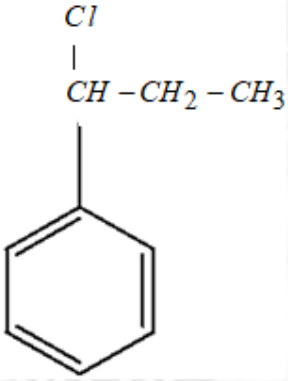
1)



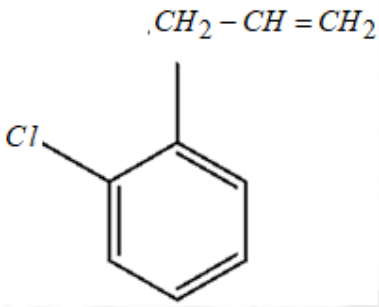
2)



3)



4)



59. Which of the following order is/are correct?

1) $SbH_3 > NH_3 > AsH_3 > PH_3$- Boiling point

2) $NH_3 > PH_3 > AsH_3 > SbH_3$-Reducing strength

3) $NH_3 > PH_3 > AsH_3 > SbH_3$-Basic Character

4) $NH_3 > PH_3 > AsH_3 > SbH_3$-Bond Angle

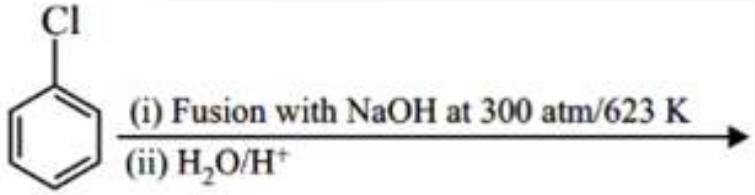
1) 1, 2 and 3 only **2)** 2, 3 and 4 only **3)**1,3 and 4 only **4)** 1,2, 3 and 4

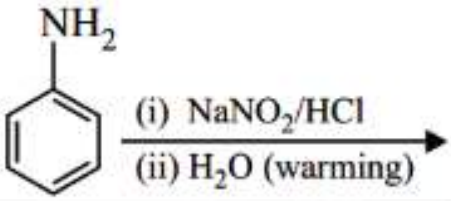
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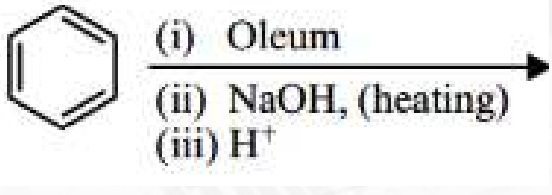
60. Passing H_2S into a mixture of Mn^{2+}, Ni^{2+}, Cu^{2+} and Hg^{2+} ions in an acidified aqueous solution precipitates

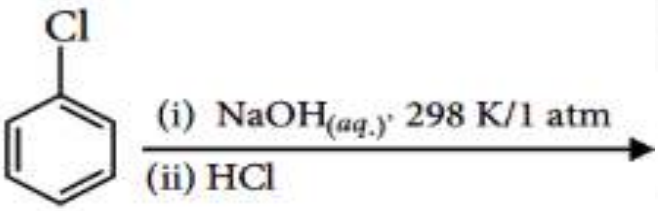
1) MnS and CuS **2)** CuS and HgS **3)** NiS and HgS **4)** MnS and NiS

61. Which of the following reactions will not yield phenol?

1) 

2) 

3) 

4) 

62. Two isomeric ketones, 3-pentanone and 2-pentanone can be distinguished by:

1) I_2 / NaOH **2)** $NaHSO_3$ **3)** *NaCN / HCl* **4)** 2, 4-DNP

63. $CrCl_3\ xNH_3$ can exist as a complex, 0.1 molal aqueous solution of this complex shows a depression in freezing point of 0.558^0C. Assuming 100% ionization of this complex and coordination number of Cr is 6, the complex will be

(Given K_f of water = 1.86 K kg mol^{-1})

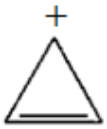
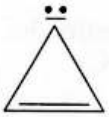
1) $\left[Cr(NH_3)_6\right]Cl_3$ **2)** $\left[Cr(NH_3)_4Cl_2\right]Cl$

3) $\left[Cr(NH_3)_5Cl\right]Cl_2$ **4)** $\left[Cr(NH_3)_3Cl_3\right]$

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64. Which of the following structure is aromatic in nature?

1)  **2)**  **3)**  **4)** 

65. Bond enthalpies of H_2, X_2 and HX are in the ratio 2:1:2. If enthalpy of formation of HX is $-50 kJmol^{-1}$, the bond enthalpy of H_2 is?

1) 300 $kJmol^{-1}$ **2)** 100 kJ mol^{-1} **3)** 200 kJ mol^{-1} **4)** 400 kJ mol^{-1}

66. Given below are two statements: one is labeled as Assertion (**A**) and the other is labeled as Reason (R).

Assertion (**A**): Haloalkanes react with KCN to form alkyl cyanides as a major product while with AgCN form isocyanide as the major product.

Reason (R): KCN and AgCN both are highly ionic compounds.

In the light of the above statements, choose the most appropriate answer form the options given below.

1) Both (**A**) and (R) are correct and (R) is the correct explanation of (**A**).

2) (**A**) is not correct but (R) is correct.

3) Both (**A**) and (R) are correct but (R) is not correct explanation of (**A**)

4) (**A**) is correct but (R) is not correct.

67. Given below are two statements:

Statement I: Ozonolysis followed by treatment with Zn, H_2O of cis-2-butene gives ethanol

Statement II: The product obtained by ozonolysis followed by treatment with Zn, H_2O of 3, 6-dimethyloct-4-ene has no chiral carbon atom.

In the light of the above statements, choose the correct answer from the options given below

1) Both Statement I and Statement II are true

2) Statement I is true but Statement II is false

3) Statement I is false but Statement II is true

4) Both statement I and Statement II are false

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68. Given below are two statements; one is labeled as Assertion (**A**) and the other is labeled as Reason (R).

Assertion (**A**) : NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3.

Reason (R): In NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N-H bonds. F is the most electronegative element. In the light of the above statements, choose the correct and answer from the options given below:

1) (**A**) is true but (R) is false

2) Both (**A**) and (R) are true (R) is correct explanation of (**A**).

3) Both (**A**) and (R) are true but (R) is not correct explanation of (**A**)

4) Assertion (**A**) is false but Reason (R) is true.

69. Statement I: Cr^{2+} *and* Mn^{3+} both have d^4 configuration.

Statement II: Cr^{2+} is an oxidizing agent while Mn^{3+} is a reducing agent

1) Both statement I and Statement II are correct

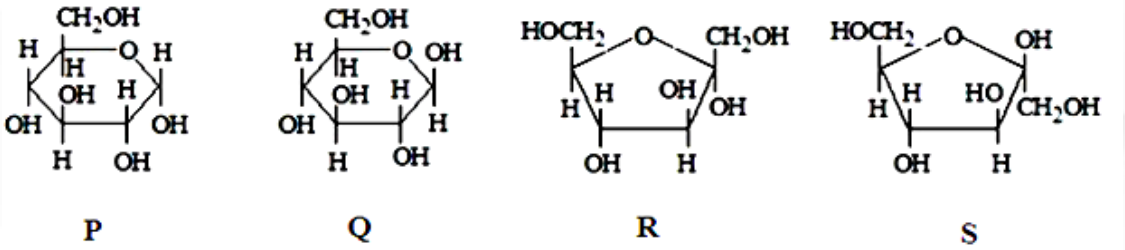
2) Statement I is correct but Statement II incorrect

3) Statement I is incorrect but Statement II is correct

4) Both Statement I and Statement II are incorrect

70. Match items of Row I with those of Row II

Row I :



P Q R S

Row II:-

(i) α-D-(-)- Fructofuranose (ii) β-D-(-) - Fructofuranose

(iii) α-D-(-)- Glucopyranose (iv) β-D-(-)- Glucopyranose

Correct match is :

1) P→iii;Q→iv;R→ii;S→i **2)** P→iii;Q→iv;R→i;S→ii

3) P→iv;Q→iii;R→i;S→ii **4)** P→i; Q→ii; R→iii; S→iv

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SECTION-II

(NUMERICAL VALUE TYPE

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Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases

71. For an electrochemical cell, $Sn(s)\left|Sn^{2+}(aq,1\ M)\right\|Pb^{2+}(aq,1M)\left|Pb(s);\right.$ the value of $\log\frac{\left[Sn^{2+}\right]}{\left[Pb^{2+}\right]}=\frac{1}{x}$ when this cell attains equilibrium. Then x=

$\left(Given: E^{0}_{Sn^{2+}/Sn}=-0.14V; E^{0}_{Pb^{2+}/Pb}=-0.13V, \frac{2.303RT}{F}=0.06\right)$

72. 29.2% (w/w) HCl stock solution has a density of 1.25 g/mL. The molecular weight of HCl is 36.5 g/mol. The volume (mL) of stock solution required to prepare a 200 mL solution 0.4 M HCl is:_____

73. Number of lonepairs in I_3^- _____

74. How many of the following are optically inactive?

i) trans -$\left[Co(en)_2Cl_2\right]^{2+}$ ii) cis- $\left[Co(en)_2Br_2\right]^{2+}$ iii) $\left[Co(NH_3)_3Cl_3\right]$

iv) trans-$\left[Co(NH_3)_4Cl_2\right]^{+}$ v) trans-$\left[CoCl_2(C_2O_4)_2\right]^{3-}$

75. In an atom, the total number of electrons having quantum number $n=4, |m_l|=1$ *and* $m_s=-\frac{1}{2}$ is

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